

Concourse: The Event Loop

Concourse research notes

July 17, 2026

Contents

1	What this document is	1
2	D1: fire emits clock deltas	1
2.1	Why the diff loses	1
2.2	The delta vocabulary	1
2.3	The debug oracle	2
2.4	Refinement forced by F2: deltas are derived, not emitted	2
3	D2 proposal: the cascade worklist	3
3.1	The problem	3
3.2	Split contention from propagation	3
3.3	What falsifies it	4
4	D3: the interpreter owns the record	4
5	D4: the builder chooses the sampler	4
6	The interpreter, whole	4
7	Phase 2: the processor-sharing compilation (F8)	5
8	Phase 2: the branchable world	6
9	Charter extensions	7
10	The Question 3 experiment: IPA over processor-sharing records	8

1 What this document is

`model_definition.tex` §8 queued four middle-layer questions. Design discussion on July 17, 2026 settled three and framed the fourth:

D1 (decided). `fire` emits enable/disable/re-enable deltas; the full `enabled(state)` recompute is retained as a debug-mode consistency check, not measured against — deltas are chosen for correctness of the order of operations, not for speed.

D2 (open; proposal in §3). The cascade worklist must settle to a fixed point; the policy that makes the fixed point unique is proposed here and awaits a decision.

D3 (decided). The interpreter owns the marked record. `CompetingClocks` samples, and only samples; its watchers observe sampling, and the marked record is not a sampling artifact.

D4 (decided). Sampler choice belongs to `CompetingClocks`' `SamplerBuilder`: `Concourse` describes the clock-key type and the distributions in play, and the builder picks. Any sampler valid for those distributions must work; none is load-bearing.

Charter claims F9–F11 (§9) extend the falsification charter of `model_definition.tex` §7.

2 D1: fire emits clock deltas

2.1 Why the diff loses

Beyond ordering, an enabled-set diff is *semantically blind to re-enablings*. When processor sharing re-evaluates every resident clock on an arrival, or a preemption changes which distribution a still-enabled key carries, the enabled set before and after the firing is identical — the diff sees nothing, while the sampler must receive `reenable!`. Deltas carry information the diff cannot express. The diff survives only as the debug oracle for the one thing it can check: membership.

2.2 The delta vocabulary

The contract's `fire` is θ -free and pure, so deltas cannot carry distributions — only keys and operations. The interpreter derives each distribution from `clock_distribution` at the post-firing state and each anchor from `st.te` (amendment A1):

```
const ClockDelta = Tuple{Symbol,ClockKey}    # (:enable / :disable / :reenable,
      key)
# fire returns (new_state, deltas::Vector{ClockDelta}); emission order is
# authoritative and is applied to the sampler verbatim.
```

This is `ClockGradients`' existing `fire_changes(model, state, key)` seam with the changes type made concrete (and the CD-1 time argument applied to it as well); `enabled_update` consumes the same deltas, so the estimators' incremental path and the interpreter share one vocabulary. The fired clock itself is implicit: the interpreter calls `fire!(ctx, key, t)` before applying deltas, so `fire` never emits a delta for its own key.

2.3 The debug oracle

After every firing, debug mode asserts two things:

1. *Membership:* `enabled(m, st')` equals the previous enabled set plus emitted `:enables` minus emitted `:disables` (and the fired key). This is the complete-recompute check, and it is the specification — `enabled` remains the meaning, deltas the implementation.
2. *anchors:* the sampler's per-clock enabling times equal `st.te` — the A1 invariant. This is what checks re-enablings, which membership cannot see.

2.4 Refinement forced by F2: deltas are derived, not emitted

Running charter claim F2 (add reneging touching only the family registry and the surface constructor) falsified D1 *as first implemented*: patience clocks are born when a job enters a buffer and die when it leaves, and with deltas hand-emitted inside `settle!` and the family `fire` functions, adding the family would have meant threading emission code through every state-transition site — interpreter surgery, which is exactly what F2 forbids. The repair keeps D1’s decision and changes its mechanism:

- *Membership deltas are derived, not emitted.* The cascade already knows which stations it touched. After the cascade, for each touched station and each family, the family’s per-station enabled keys are diffed between the old and new states (both exist; `fire` is pure). A family is now exactly four methods — `family_station_keys!`, `family_law`, `family_fire!`, `family_memory` — and clock lifecycle follows from membership alone.
- *Locality is preserved.* The user’s original argument for deltas over a global diff was cost on large networks; the derived diff runs only over the touched stations’ clocks, so the asymptotics stand. Explicit `:reenable` deltas remain available for the processor-sharing path (phase 2), which membership cannot see.
- *The A1 bookkeeping collapses into one place.* Enabling times and banked ages are updated in the same derivation pass, driven by the diffs and the per-family memory policy, instead of at every site that moves a job.
- *A correctness dividend.* A clock enabled and disabled within one firing (a job dispatched and immediately preempted in the same cascade) now nets out of the diff and never reaches the sampler — the GSMP has no zero-duration clocks, which the emission order previously only approximated by a disable/enable pair.

This is the falsification charter working as designed: the claim failed against the first implementation, and what survived is a better mechanism under the same decision. After the repair, the `:patience` addition itself changed only the registry and the surface fields, and $M/M/1+M$ matches the Erlang-A birth–death form with the membership oracle running through every reneging cascade.

3 D2 proposal: the cascade worklist

3.1 The problem

Within one firing at time t , instantaneous consequences chain: a deposit triggers `settle!` at the destination; a departure from a full buffer lets a blocked upstream server transfer in, which frees that server, which admits from its buffer, which may unblock stations further upstream. The cascade must reach a fixed point — no station can dispatch, preempt, or unblock — before the sampler sees any delta. The requirement is that the fixed point be *unique and deterministic*, because `fire` is the replayed, differentiated object.

The GSPN literature has this exact problem as immediate transitions and vanishing markings, resolved there by priorities and weights (Zimmermann Ch. 5–6). The proposal below is that resolution, restated for queues.

3.2 Split contention from propagation

The order of cascade processing matters in exactly one place: when two consumers compete for one freed resource — two blocked upstream servers and one freed buffer slot; two idle servers and one arriving job is already settled by the discipline’s `select`. Everywhere else the steps merely propagate.

Contention is model semantics. Who gets a freed slot is an observable modeling choice, not an implementation detail, so it must be a declared policy value on the station, parallel to the discipline:

```
UnblockPolicy:
  FCFSUnblock()           # longest-blocked first (default)
  PriorityUnblock(by)      # by a ScalarExpr over the blocked job's marks
  RandomUnblock()         # a recorded choice, via the A4 draw source
```

`FCFSUnblock` needs the blocking order, so the blocked-server set in `QueueState` is an ordered vector, not a set. `RandomUnblock` consumes a recorded draw and so stays inside the A4 randomness census.

Propagation must be order-free. With contention delegated to policy, the claim is that the remaining cascade relation is *confluent*: any fair processing order reaches the same fixed point. Termination comes from check C3 (acyclic blocking chains): each unblocking transfer moves a job strictly downstream in the blocking DAG, so the cascade measure is well-founded. Confluence plus termination give a unique normal form (Newman’s lemma); the worklist order is then an implementation choice, not a semantic one.

The canonical order, so that the implementation is deterministic even if the confluence claim fails: a FIFO worklist of stations to settle, seeded with the destination station then the origin station of the firing — the one-hop order `queue_layers.tex` §3.2 fixed by fiat, generalized. A debug-build fuel counter bounds the cascade at (jobs + servers) steps and turns nontermination into an error rather than a hang.

3.3 What falsifies it

F9 in the charter: run the same trajectories with FIFO and LIFO worklists; `states_equal` at every index or the confluence claim is dead, and the worklist order gets promoted into the IR’s semantics the way the one-hop order already is. Either outcome is a result; the test is cheap.

4 D3: the interpreter owns the record

The marked record (k_i, t_i, a_i) is the primary observable of the whole library, and its auxiliary-draw column exists because the *interpreter* supplies the draw source (A4) — the sampler never sees those draws, so a sampler-side watcher could never capture them. The interpreter appends one entry per firing: the key, the time, and the draws consumed by that firing in consumption order. CompetingClocks’ watchers remain what they are — instrumentation of sampling — and Concourse attaches none of the record path to them. `gradient_record` gets one method, from the marked record type, per `model_definition.tex` §6.

One replay detail follows from record ownership: during estimation, **fire** must read firing *i*'s recorded draws, and the contract does not pass *i*. **QueueState** therefore carries a firing counter (**nfired::Int**, incremented by every **fire**), and the replay binding indexes the record with it. A pure counter in a pure state; the same device as **next_id**.

5 D4: the builder chooses the sampler

Concourse tells **SamplerBuilder** the clock-key type and lets **CompetingClocks** choose the sampler for the distributions in play; the default is whatever the builder decides, and Concourse hard-codes no method. The design consequence is a claim, not a configuration: *every* sampler valid for the model's distribution set must produce statistically identical results, because the interpreter speaks only the blessed **SamplingContext** surface (**enable!**, **disable!**, **reenable!**, **next**, **fire!**) and no model code reaches around it. Estimator-driven overrides (the carry coupling for IPA) are passed through as builder arguments by the estimation entry points, not chosen by the model.

6 The interpreter, whole

With D1–D4 fixed, the middle layer is small enough to state completely. Julia-flavored pseudocode; error handling and the debug oracle elided:

```
function simulate(m::QueueGSMP,  $\theta$ , horizon, seed)
    st = initial_state(m)
    ctx = SamplingContext(build(m), seed)      # D4: builder chooses
    rec = MarkedRecord(clockkeytype(m))
    for k in enabled(m, st)                    # initial enablings
        enable!(ctx, k, clock_distribution(m,  $\theta$ , k, st), st.te[k], 0.0)
    end
    while true
        (t, key) = next(ctx, st.time)
        (key === nothing || t > horizon) && break
        fire!(ctx, key, t)                     # realize the winner's draw
        draws = draw_source(ctx, key)          # A4: keyed, recording
        st, deltas = fire_changes(m, st, key, t, draws) # cascade inside
        push!(rec, key, t, consumed(draws))    # D3: interpreter-owned
        for (op, k) in deltas                  # D1: emission order verbatim
            op === :disable && disable!(ctx, k, t)
            op === :enable && enable!(ctx, k, clock_distribution(m,  $\theta$ , k, st),
st.te[k], t)
            op === :reenable && reenable!(ctx, k, clock_distribution(m,  $\theta$ , k,
st), st.te[k], t)
        end
    end
    close_record!(rec, min(horizon, st.time))
end
```

Everything model-specific is inside **fire_changes** (families, disciplines, kernels, the cascade); everything stochastic is inside **ctx** and **draws**; everything observable is **rec**. The loop itself should never grow.

7 Phase 2: the processor-sharing compilation (F8)

Implemented July 17, 2026. Three findings, and one normative rule the implementation surfaced, stated first because it governs the rest.

The declaration rule (normative). A model’s entire specification for an enabled clock is the pair (F, t_e) , re-declarable only at event times. Its meaning is fixed by the conditional law: given the history up to the current time t , the firing time is distributed as $t_e + X$, $X \sim F$, conditioned on $t_e + X > t$. Nothing else crosses the interface — not a speed, not an age, not how the sampler realizes the draw (retention and coupling are the sampler’s construction-time properties; the context forwards neither). “Speed” is a modeling-side device used to *construct* the declared F ; per `queue_layers.tex` decision #1 it compiles away, and here it did: **bank** and **anchor** never reach the sampler. Two declarations with the same conditional law are the same model. In particular, at a PS change at time t^* with internal age a and new rate r ,

$$(F_A, t^*), \quad F_A = \frac{X-a}{r} \mid X > a \quad \text{and} \quad (F_B, t_e), \quad F_B = (t^* - t_e) + \frac{X-a}{r} \mid X > a$$

both put the firing at $t^* + (X-a)/r$ given $X > a$: the anchored-at-now form of `queue_layers.tex` §3.5 and the fixed-anchor form implemented here are the *same conditional law*, and the F8 test “the two anchor parameterizations declare the same conditional law” pins the identity numerically. The choice between them is therefore not semantics but *which sufficient statistics the state records*, and that choice is forced by the replay bookkeeping: the ClockGradients Bookkeeper reconstructs t_e from the enabled rule and holds it fixed while a clock stays enabled, the score replay accumulates piecewise survival against distributions at that fixed anchor, and the segment machinery (CG-M3) represents exactly re-declarations at constant t_e .

The state carries (a, anchor) , an A1 addendum. The implemented declaration is the fixed-anchor F_B above, built from **bank** (internal age a at the last re-declaration) and **anchor** (that re-declaration’s wall time). The sampler side is the age-keeping **reenable!** (**relative_te** = **te** – **now** ≤ 0), which CompetingClocks documents as exactly this case. The interpreter did not change: the **:reenable** branch it already had for D1 computes that shift.

The re-evaluation hook. The derived-delta pass (§2.4) gained its promised explicit-**:reenable** seam: **family_reenables!**, a per-family hook run after the membership diff at each touched station. The **:service** method detects a resident-count change at a PS station, banks each survivor’s internal age at the old speed, moves its anchor, and emits (**:reenable**, **k**). A **:rescale** memory mode joins **:fresh**/**:resume**: enabling anchors **te** at now and clears the bank, and speed bookkeeping otherwise stays out of the generic passes.

Dual-safety forced a bespoke distribution. Composing `Distributions.truncated` and affine wrappers builds the right law but poisons ForwardDiff: `Truncated`’s stored bound constants produce NaN partials in `logccdf` (and a lower bound of zero stores $\log 0 = -\infty$ with NaN partials outright). `SharedRemaining` in `src/semantics.jl` implements the conditional-remaining law directly — every method a two-call delegation to the base distribution — and is the object the sampler draws from, the likelihood differentiates, and the Bookkeeper compares for segment detection. (The hunt also surfaced a one-line short-circuit-order bug in CompetingClocks’ `lefttrunc.jl`, fixed there.)

The oracles: M/M/1-PS matches $\rho/(1-\rho)$; M/G/1-PS with Gamma service is *insensitive* (same mean, while the same law under FCFS matches its distinct Pollaczek–Khinchine mean — proving the discipline actually shares); replay reproduces **bank**/**anchor** exactly; every resident-count change emits re-enables for every survivor; and the score gradient matches finite differences

through genuine segment chains — the first exercise of CG-M3’s mid-flight machinery by a real model, and the validation of the capability table’s “likelihood ✓” row for PS.

8 Phase 2: the branchable world

Implemented July 17, 2026: `ConcourseWorld = (QueueGSMP,  $\theta$ , QueueState, SamplingContext, time)` implements ClockGradients’ ten-verb branchable protocol — the second implementation beside the ChronoSim adapter, the same two-clients test the model contract passed. The verbs are thin: `peek` is the context’s cached reservation; `commit` and `force` share one update path (`fire!/force_fire!` on the context, then `fire_changes` plus the interpreter’s own `apply_deltas!`); a coupled clone is `empty-clone + copy_clocks!` + time restore; `rekey` is `rekey-then-jitter!` (reseeding alone would replay cached putative times); ages come from the state’s own `te` table (the A1 invariant makes the sampler agree), sorted by key; the dual- θ distribution rebuild is `clock_distribution` at the world’s state, with `SharedRemaining` delegating `partype` to its base so the dual probe sees θ .

Auxiliary draws (the Q1 decision, implemented July 17, 2026): the world owns a live `KeyedStreams` family beside the sampler, and `fire` runs from it, so models with marks, probabilistic routing, and random selection branch too. The stream key is the stable identity the draw belongs to. A mark or source-routing draw is keyed by the source’s own clock and consumed once per arrival, so the k -th value *is* the k -th arrival of that source in any same-seed world, whatever else fired in between — the “position in the source’s own sequence” rule. A station’s routing or selection draw is keyed by the job’s clock and stays aligned exactly when job identities do. The three obligations are tested: a clone copies the generators’ states (a coupled clone reproduces firings and marks exactly), a rekey re-seeds the auxiliary family with the clock family (fresh sizes, not replays), and two same-seed clones agree on the marks of corresponding jobs even after a forced firing makes their event orders diverge. θ -dependent mark laws stay refused at the factory: a mark whose law reads θ adds its own derivative term to every estimator, and that term is future work, not a default. Because marks and routing weights are θ -free, their likelihood factors are constants in θ and the score estimator needs no term for them. The live model binding (`live_model`) still gives SPA’s twin an off-record `fire` that fails loudly if a draw is requested, because the twin has no streams of its own.

CD-1’s last estimator threading landed with this work: SPA’s commuting gates and forced twins fire at the swap time, the horizon jump fires at the horizon, and `fire_changes` gained its own four-argument default in the contract.

Validation: `check_branchable` passes in full for both an M/M/1 world and a PS world (clone coupling and rekey divergence across mid-flight re-declarations); `branching_gradient` matches finite differences on the terminal number-in-system for M/M/1 *and* for PS — the capability table’s “needs framework world” cell, now green for the branching estimator; `spa_gradient` matches finite differences on M/M/1 occupancy; and `branching_gradient` matches finite differences on the bernoulli split (routing draws) and the SITA split (mark draws) through the world-owned streams. SPA on PS remains refused upstream by its own multi-segment guard, and IPA over PS records is now refuted outright (§10) — both wait on the Question 2 derivation.

9 Charter extensions

Continuing the numbering of `model_definition.tex` §7:

- F9 (cascade confluence).** With contention resolved by `UnblockPolicy`, the cascade fixed point is order-independent. *Test:* FIFO vs. LIFO worklists over random finite-buffer `:block` networks; `states_equal` at every firing index. Falsified $\Rightarrow$ the worklist order becomes IR semantics.
- F10 (sampler agnosticism).** No charter result depends on the sampler. *Test:* run the F1–F7 oracle comparisons under at least two builder outcomes (e.g. first-to-fire and next-reaction); all pass under both.
- F11 (delta fidelity).** Emitted deltas agree with the recompute oracle. *Test:* debug-mode membership and anchor assertions (§2) hold over long mixed runs including preemption — and, in phase 2, processor sharing, which is the case only the anchor check can see.
- F12 (protocol portability).** The branchable protocol is an abstraction, not a description of one adapter. *Test:* `check_branchable` passes in full on `ConcourseWorld` for a frozen-law model and for PS (re-declarations under clone/rekey).
- F13 (clone estimation over re-declared clocks).** The branching estimator is valid when clocks are re-declared mid-flight. *Test:* `branching_gradient` vs. common-random-number finite differences on the PS terminal count, within 4 combined SEs.
- F14 (pathwise estimation over re-declared clocks).** *Refuted; see §10.* The claim was: the frozen-order IPA replay, which was written with multi-segment records in mind, gives the correct derivative on processor-sharing records. The experiment says no, under both couplings. *Test:* the refutation is pinned in `test_ps.jl` with fixed seeds, alongside the mechanical facts that survived (duals flow through `SharedRemaining`; the replay reproduces the recorded times at the sampling parameter).

10 The Question 3 experiment: IPA over processor-sharing records

The design brief asked whether IPA, run over a processor-sharing record, matches finite differences. The experiment (M/M/1–PS, $\lambda = 1$, $\mu = 2$, horizon 50, 400 records, common-random-number central differences on the same seeds, functional $\int_0^H N dt$) answers no:

	∂_λ	∂_μ
finite differences	93.4 ± 3.9	-47.7 ± 1.8
IPA, redraw coupling	57.7 ± 3.4	-10.0 ± 0.1
IPA, carry coupling	206.0 ± 11.8	-36.2 ± 1.2

(The steady-state truths are +100 and −50 times the horizon fraction at stationarity; the finite differences sit where the transient predicts.) Three facts sharpen the failure into something usable.

First, the duals flow without errors. `SharedRemaining` was admitted to the dual-safe gate by the recursive rule (a wrapper is dual-safe when its base law is), no exceptions surfaced, and the Float64 replay reproduces the recorded firing times. The failure is a wrong number, not a crash.

Second, the bias is attributable to the record’s representation of segments, not to the dynamics. The sharpest evidence is a degenerate experiment: a processor-sharing station with 64 servers re-declares every survivor at every count change, yet the speed is always 1, so the dynamics are

exactly those of the same station under FCFS — but the PS run’s records carry multi-segment chains while the FCFS run’s records are segment-free. On the segment-free records both couplings match finite differences. On the multi-segment records of the *same law* both are badly biased (redraw ∂_μ : -29.5 vs the true -143 ; carry ∂_λ : 233 vs the true 49), and the per-path carry gradients differ from their segment-free twins by up to several hundred. Identical dynamics, opposite verdicts: the segment representation is the whole story.

Third, the mechanism is located, and it is structural rather than a one-line defect. The replay recurrences rebuild each segment’s law from states folded at the *recorded* Float64 times, so the law’s **bank**, **anchor**, and speed are frozen constants; and the conditional-survival term that re-anchors a re-declared clock is evaluated exactly at the segment’s start, where the declared law’s support begins and the term is pinned to zero whatever the boundary time’s derivative is. Both facts say the same thing: the derivative of a re-declaration *boundary time* — itself a θ -dependent event time of the race being differentiated — has no channel into the replayed times. (A candidate one-line repair, taking the formula branch at the support boundary, was tried and changed nothing: round-off places the evaluation on either side of the boundary, and the frozen anchors remain either way.) A complete repair needs the segment parameters to be dual-valued functions of the replayed times, which is Question 4’s territory — either the reconstruction learns to fold at dual times, or the record carries the segment boundaries in a differentiable form. What the correct derivative even is, once that channel exists, is Question 2’s derivation.

Consequence for users today: on processor-sharing records the valid estimators are the score function and branching (both tested against finite differences); IPA silently returns a confident wrong number, which is the failure mode the score pairing exists to detect. The refusal question — whether the IPA entry point should reject multi-segment records the way SPA already does — belongs to the Question 2 derivation, because the derivation will say which multi-segment records are safe rather than banning them all.